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Highlights

- Accelerated resin degradation of a Protein A chromatography column was studied
- Various data analysis methods were used on chromatogram transitions
- Traditional methods like HETP and asymmetry did not show any consistent trends
- PCA, PLS, Similarity Score, SOP-MPT systematically track changes in resin integrity
- Proof of concept shown here, further work for validation as a PAT tool is required

Exploring features in chromatographic profiles as a tool for monitoring column performance

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Abstract

In the biopharmaceutical industry, chromatography resins have a finite number of uses before they start to age and degrade, typically due to losses of ligand integrity and/or density. The "health" of a column is predicted and validated by running multiple cycles on representative scale-down models and can be followed by real-time on-going validation during commercial production.

Principal Component Analysis (PCA), Partial Least Square (PLS), Similarity Scores and Single One Point-MultiParameter Technique (SOP-MPT) along with machine learning principles were applied to explore the hypothesis that there is predictive capability of latent variables in chromatography absorbance profiles for process performance (step yield) and product quality (aggregates, fragments, host cell proteins (HCP) and DNA, and Protein A ligand).

The first stage of this study is described in this paper: a MabSelect SuRe™ chromatography column was cycled with a method to establish the "normal" baseline for process performance and product quality, followed by runs using a harsher NaOH Cleaning in Place (CIP) procedure (with a higher NaOH concentration than that recommended by the vendor) to accelerate resin degradation. The different mathematical analytical tools correlated with resin degradation of the column (reflected in decreasing step yield and binding capacity with increasing running cycle), specifically when using the Wash, Elution and Strip phases of the chromatography method. Monomer, HCP and DNA content were not significantly impacted and therefore a correlation with product quality was inconsequential. Importantly, this work shows proof-of-concept that while more traditional methods of measuring resin integrity such as the height equivalent to a theoretical place (HETP) and Asymmetry (As) measurements could not detect changes in the integrity of the resin, PCA, PLS, Similarity Scores and SOP-MPT (to a lesser extent) applied to the absorbance data were capable of anticipating issues in the chromatography bed by identifying atypical outcomes.

Keywords

PAT, chromatography, resin degradation, proof-of-concept modeling

1. Introduction / Background

Monitoring and predicting resins' lifetime before performance and product quality issues arise can reassure purification performance through extended number of runs. Specifically, tracking column health in real time will allow a prompt process control response to various potential problems. This is especially important during continuous processing, when the full lifetime of resins is more likely to be met.

Traditionally, resin lifetime is determined through extensive cyclic chromatography runs on a representative scale-down model under typical running conditions, through inspection of chromatographic profiles and off-line analytical assays whose results may take hours to days. This exercise can also be a part of on-going validation at the commercial production scale. Height equivalent to a theoretical place (HETP), asymmetry (As), and Transition analysis from shifts in salt or pH during a chromatography run are commonly used to determine and monitor column bed integrity [1, 2]. However, these practices do not capture what is happening to the column in real time and at the ligand level. Thus, column degradation may not be noticed until deviations in process performance or product quality arise later down the line.

While resin aging is still not completely understood at a fundamental level, it is believed to occur by three main mechanisms [3]. First, the binding capacity of the resin can be decreased through ligand leaching, occlusion or inactivation, which can lead to reduced process performance or impact to product quality. An alternative mechanism is related to the mass transfer rate that may be affected by breaches to column packing integrity. Furthermore, fouling of the resin caused by inadequate cleaning [4] can lead to carry-over between cycles, potentially impacting product quality. There are several studies described in literature around the understanding of resin aging that often include sophisticated process analytical technologies (PAT) and modelling approaches [4, 5, 6]. Other studies of column aging have been performed on historical data collected under typical operational conditions. Such retrospective studies are valuable but may be less effective in identifying the best method for column monitoring since typically no major deviations are present in the dataset.

Specifically, being able to identify atypical aging of protein A affinity chromatography resins is of significant importance. These resins are widely used for capturing monoclonal antibodies, but they are expensive and have a limited lifetime relative to the ion exchange-based resins due the proteinaceous nature of the ligand. This paper uses a protein A resin as a model to understand if resin degradation can be inferred by investigating step-wise modifications in tracelines for selected chromatography phases for absorbance, pH and conductivity. The study used a column whose integrity was challenged through a Cleaning in Place (CIP) strategy that is referred to in this paper as “harsh CIP”. This harsh CIP used a NaOH concentration above that recommended by the vendor to intentionally degrade the resin faster [7].

Principal component analysis (PCA) [8], Partial least squares (PLS) [9], Similarity Score [10, 11] and the Simple and One-Point Multiparameter Technique (SOP-MPT) [10, 11] were mathematical tools tested to determine features in the profiles that correlate with column health. The chromatogram was divided into phases and data analysis tools were mainly applied to information-rich phases, characterized by absorbance signals that were under-saturated and changed with cycle number.

PCA is a general multivariate data analysis technique that is used for dimensionality reduction and pattern recognition in highly correlated data sets. Most PCA studies of chromatographic data have focused on differences between samples, e.g. n-alkanes [12] , peptide mapping [13] or biological samples [14]. In such cases it is highly beneficial to preprocess the chromatographic data to reduce the influence of column degradation and mobile phase variability [15, 16, 17, 18]. However, in the current study, the chromatogram preprocessing is focused on enhancing the ability to detect column aging. Examples of more bioprocess-related applications of chromatogram PCA include studies of influential factors affecting chromatogram shape [19, 20] and for separation modeling [21]. There are some studies that look at PCA and related techniques as a way to monitor column performance. The early work of Larson et al [1] included a brief but forward-looking study of the utility of

PCA for column monitoring. More recent examples that include relationships to product quality data have been presented by Hou et al [22], Gerberich et al [23] and Feidl et al [24]. These works demonstrated the potential of PCA-based methods to model resin aging and detect column integrity failures, as well as to predict column performance indicators such as yield, purity, and dynamic binding capacity.

PLS regression is a statistical method that, similarly to PCA, can be used for dimensionality reduction. Unlike, PCA, PLS is a supervised method which requires a target (dependent) variable. PLS model tries to find the multidimensional directions in the feature space that explain the maximum variance in the target space. PLS is commonly used for conversion of a sensor signal to product quality data in PAT applications [25, 26] and to construct Quantitative Structure Retention Relationships (QSRR) [27, 28]. PLS has also been used to predict step yield and other quality attributes of the process such as HCP and DNA content [24]. In this work, we used PLS to predict the cycle number, which is similar to the approach taken for batch modeling in the SIMCA software [29] where the input signals are used to model and predict the batch time.

The Similarity Score [10, 11] uses two measures calculated for relevant phases (here the Elution phase) in the chromatogram: the Pearson correlation coefficient between a reference chromatogram and each chromatogram under investigation, and the maximum signal level within the phase for each chromatogram.

SOP-MPT is defined as collecting data from multiple data types/curves (e.g. pH, conductivity) at the point of the absorbance peak max (if non-saturated). The intention is to highlight deviations from the correlation pattern between the different curves in the chromatogram as the column degrades [30].

The analysis presented in this paper is intended to be a proof-of-concept to demonstrate that chromatographic profiles can be used to predict the state of a column as it relates to the integrity of resin. While the use of such monitoring strategies may not explain the cause for atypical chromatography behavior, it can support a robust control strategy based on the understanding of how chromatography profiles are linked to process outputs. The use of PLS, PCA, Similarity Scores and SOP-MPT on at least one of the phases *Wash*, *Elution* and *Strip* of the chromatography method showed good correlation with column age. Therefore, already existing output measurements collected in routine chromatography steps have the potential to be used as PAT tools without the need for additional sensors.

2. Materials and Methods

2.1. Chemicals, monoclonal antibodies, and Protein A resins

Buffer preparation used chemicals obtained from JT Baker (Radnor, Pennsylvania, USA). The antibody was a human IgG_1 provided by AstraZeneca (Gaithersburg, MD, USA). The protein A resin used in this project was MabSelect Sure™ (MSS), supplied and packed according to instructions provided by Cytiva (Marlborough, MA, USA).

2.2. Column Hardware, chromatography Instrument and Methods

An Omnifit® column (Diba Industries, Danbury, CT, USA) was used for all experiments in this study. It had an internal diameter of 0.66 cm and was packed to a bed height of 21.7 cm, which resulted in a column volume of 7.4 ml. An AKTA Avant, using Unicorn software version 7.1, was used to carry out the chromatography runs (Cytiva, Marlborough, MA, USA). A total of 30 chromatography runs were performed. Briefly, the “normal” chromatography (first 15 runs) used 8 minute residence times for all phases and 12 minute residence time for the load phase. All runs were performed at room temperature. Column was equilibrated with a Tris buffer at pH 7.4, loaded to 50 mg/ml resin, washed with a high NaCl containing buffer at the same pH, and eluted at pH 3.6 in acetate buffer. After each run, the resin was stripped with 50 mM glycine pH 2.3 and was then cleaned in place with 0.1 N

NaOH using a static hold of 35 min upon equilibration in base. Runs with harsh CIP used 1 N NaOH instead of 0.1 N NaOH in subsequent 15 runs, with similar contact time.
Conductivity and pH were monitored by the in-line probes of the AKTA system. Protein concentration of the product pools were measured off-line using a NanoDrop™ 2000 (Thermo Fisher Scientific, Wilmington, DE, USA).
Six measurements of HETP and As (pulse measurements) were performed throughout the study [7] using 100 mM to 1 M NaCl. Samples of the product were also aliquoted and analyzed for purity, host cell protein and DNA. The acceptance criteria for columns to pass qualification was an asymmetry value between 0.8 and 1.4, and an HETP less than 0.1 cm.

2.3. Dynamic Binding Capacity (DBC) Experiment

The protein A column was loaded to 70 mg/ml at ~8 min residence time and the flow through stream was fractionated and analyzed for antibody concentration using a Protein A bindable column (Thermo Fisher Scientific, Waltham, MA, USA). Dynamic binding capacity was defined as the amount of protein loaded per volume of packed resin, for which a 10% breakthrough of the original load material concentration was reached.

2.4. Analytical methods

2.4.1. Protein concentration

Protein concentration of eluted products was measured based on the absorbance at 280 nm of the monoclonal antibody (mAb). Step yield was defined as the total mass of product by absorbance in the eluted fraction relative to the antibody mass in the cell-free cell culture loaded onto the column.

2.4.2. Host cell protein (HCP) and leached protein A (lProtA)

These impurities were measured using Enzyme-Linked Immunosorbent Assays (ELISA), with proprietary reagents.

2.4.3. Host cell DNA (DNA)

Host cell DNA was based on a quantitative PCR (qPCR) PCR assay using CHO SINE Forward (20-mer) and CHO SINE Reverse (21-mer) primers.

2.4.4. Purity

High performance size exclusion chromatography was used to measure percent monomer, fragment and aggregate. A TSK3000 column (Tosoh, King of Prussia, Pennsylvania, USA) was used to elute the various species based on molecular size.

2.5 Data analysis methods

Cycles 1, 7 and 8 were identified as outliers caused by a mismatch of the time scales during file export resulting in significantly deviating chromatogram shapes and hence excluded from data analysis when using PCA, SOP-MPT and Similarity Scores.

2.5.1. Data preprocessing

Chromatograms were exported as csv-files from UNICORN and subsequently preprocessed using an in-house script in R [31]. The preprocessing consisted of baseline subtraction in cases where substantial baseline differences occurred between cycles and volume-axis interpolation to put all curves on equidistant volume series with a common resolution of 0.1 mL. The output created segmented data sets where each curve was expressed individually for every chromatographic phase with all cycles (chromatograms) aligned to the start of the phase. This resulted in one aligned dataset per phase and curve where the most information rich datasets, as judged by visual chromatogram inspection, were analyzed separately.

2.5.2. Transition analysis

The selected salt transition from equilibration buffer to the high NaCl wash in the same buffer during the *Wash* phase (see Figure 1) was converted to a peak shape by differentiation of the conductivity curve and subsequently analyzed for HETP and asymmetry by standard peak integration methods using the UNICORN software.

2.5.3. Principal Component Analysis

The principal component analysis was performed according to standard methods implemented in SIMCA v17 (Sartorius, Gottingen, Germany). The dataset matrices were set up with the cycles as objects and the volume points as variables. Scaling was restricted to mean centering as all the variables were measured on the same scale, e.g., absorbance at 280 or 600 nm. The PCA models were determined using the first 15 cycles (excluding the outliers identified above) as the training set. The number of principal components (PCs) retained was determined from the incremental explained variance and in all cases restricted to the first two or three components, and denoted as PC1, PC2 and PC3. Results are presented as score plots including a 95% Hotelling's T2 ellipsis and distance to model (DModX) plots where the latter expresses the residuals to the model hyperplane composed by the retained principal components. The critical limit in the DModX plots (Dcrit) is calculated from the F-distribution as described in SIMCA.

2.5.4. Partial Least Squares regression

A PLS regression model was trained on the absorbance at 280 nm tracelines for *Wash*, *Elution* and *Strip* phases. A linear method is appropriate, given the relatively small number of runs available to train the model. Since PLS is a supervised method which requires a target variable, the cycle number was used as prediction goal. The model was fitted using the first 10 cycles as the training set. The analysis was performed using scikit-learn – a Python machine learning toolbox [31].

2.5.5. Similarity Score
The Similarity Score was calculated for each chromatogram in selected phases of the chromatography method. Cycle 2 was used as the reference chromatogram for the correlation part of the score. The two-dimensional plot consists of the Pearson correlation coefficient between the reference chromatogram and each chromatogram vs the maximum signal in the phase. These two metrics together cover changes both in peak shape and magnitude.

2.5.6. Simple and One-Point Multiparameter Technique (SOP-MPT)
The evaluation across curves was performed in Excel (Microsoft) by identifying the peak maximum position in the absorbance signal and reading the corresponding values for the other curves at the same volume point. The absorbance at 600 nm was used in the *Elution* phase to ensure that the signal was not saturated at the peak maximum.

3. Results / Discussion:

3.1. Chromatography results: step yield, product quality, HETP and Asymmetry
A typical protein A affinity chromatogram is shown in Figure 1. The first phase of the chromatography method comprised the resin equilibration (*Equil 1*) followed by material load (*Load*) onto the resin, a resin wash at neutral pH and high salt (*Wash*), followed by another low conductivity neutral wash (*Equil 3)* and product elution (*Elution*). The resin was then regenerated using an acidic buffer (*Strip*), followed by a phase where water for injection was applied (*WFI 1*), cleaned with NaOH (*CIP*) and re-flushed with WFI (*WFI 2*) before storage (*Storage*).
The first 15 cycles were performed with the recommended CIP conditions by the vendor [7] (0.1 N NaOH for 35 min) while cycles 16-30 used 1 N NaOH over a similar time period. Figure 2 shows the negative impact of the harsh NaOH exposure on step yield with an almost linear downward trend starting approximately at cycle 21, from about an initial range of 92-102% to a final value of 60%.
The impact on DBC after the 30 runs resulted in a decrease of about 39% relative to the original DBC: 52 versus 32 mg/ml resin (Figure 3). The steeper breakthrough curve observed after the harsh treatment of the resin corroborates the findings presented by Ling *et al* [32] suggesting that the loss of ligand is expected to lead to faster pore diffusion. The loss in dynamic capacity by the harsh CIP treatment was expected to lead to a reduction of step yield due to increasing overloading and early breakthrough during loading. The effect on step yield was estimated by integrating the area above the curve (corresponding to bound protein) in the DBC experiments suggesting a step yield of 98.6% for the fresh resin and 68.7% post harsh CIP. The experimental values were lower (see Figure 2) suggesting that additional factors impacted the step yield.

In contrast, minimal impact was found on the % aggregates, product purity, DNA, HCP (Figure 4), pH and conductivity of the eluted product (data not shown). Leached Protein A ligand was found to have a tendency to increase in the last few runs, but it was deemed to be practically negligible as all values are below 13 ppm.
Column performance was monitored using a HETP and As test, either by explicit pulse injections between cycles or based on transition analysis [1, 2] leveraging the transition of salt in the *Wash* phase of the chromatography method. The two methods for estimation gave very similar results, with a correlation of 0.97 for both HETP and As. While there was no consistent trend across the runs using “normal” versus the measurements when harsh CIP cycles were used, the HETP and As increased towards the last executed runs, indicating that the resin packing was also being negatively impacted through compression, particularly

evident around cycle 25 where the HETP abruptly increased after lowering the adaptor due to the observation of headspace formation in the column (Figure 5).

While asymmetry is systematically increasing, which is expected from a column that is incrementally being compressed (peak tailing), a sudden reduction of the asymmetry (also around run 25 - Figure 6) in the absence of any experimental changes suggests that the integrity of the packing was compromised as fronting is caused by local variations of tightness of the resin such as that observed during channeling or enough ligand has been lost [34]. In other words, data suggests that the column was increasingly being overloaded relative to the original ligand density and state of the resin. While these results may have been exacerbated by the adjustment of the top adaptor of the column before run 25, there is still an increase in asymmetry throughout the 30 cycles. In brief, it was clear that HETP and As showed similar trends (regardless of the means of measure), but they did not correlate with a gradual deterioration of column performance caused by the onset of the harsh CIP chromatography method. These observations emphasize the need for better methods for monitoring of column performance.

3.2. PCA analysis

The *Wash*, *Elution* and *Strip* were all investigated for correlation between the number of cycles and shifts in the chromatography profiles for absorbance, pH, conductivity and pressure using PCA. The tracelines for absorbance at 280 nm for the *Elution* phase were clearly outside the linear range of the detector (stated as up to 2000 mAU [35]) and thus potentially less interesting and accurate to analyze. Therefore, the absorbance signal for the elution phase was taken at 600 nm. PCA on the pH, conductivity and pressure curves did not reveal any significant patterns over the course of the study (data not shown).

All three phases - *Wash*, *Elution* and the *Strip* - visually showed a consistent correlation (positive or negative) between the progression on the number of runs and the peak height driven by the effects of the harsh CIP (Figure 7). The peak height in the *Wash* phase increases with cycle number after the harsh CIP is introduced, indicating a less tightly bound antibody and/or non-selective absorbed species. The *Elution* phase measured at 280 nm was used as pooling criterium and showed a reduction in peak width during the harsh CIP cycles that corresponds to the observed reduction in pool volume from 1.5 CV during normal CIP to 1.1 CV during the last cycle (data not shown). The peak height in the *Elution* phase measured at 600 nm is reduced in the harsh CIP cycles since less antibody is bound in each cycle when the binding capacity is increasingly lost. This is also shown by the reduction in step yield as shown in Figure 2. For the *Strip* phase, the peak height increases during harsh CIP potentially suggesting less efficient elution. The last five cycles (after lowering the adaptor) exhibit a different pattern from the remainder of the profiles which happens to be a serendipitous result that evidences the sensitivity of the absorbance profiles to events during cyclic use of chromatography columns.

The correlation was also observed graphically by PCA (Figure 8): the PCA model was calculated using the absorbance data for the first 15 cycles where three significant components covering 98% of the total variability were retained for the *Wash* phase, while both the *Elution* phase and the *Strip* phase were described by two components covering 97% (*Elution*) and 99.8% (*Strip*). Subsequently, all 30 cycles were projected onto the first two or three principal components to identify deviations from the normal pattern.

Figure 8 shows that all studied phases clearly indicate deviations in the absorbance pattern starting around cycles 17-19 and that the deviation from the signal of the first 15 cycles grows with each additional cycle number.

This is also indicated by the distance to model number, i.e., the multidimensional distance to the model hyperplane, as shown in Figure 9. Both figures also indicate a change of trend starting at cycle 26 related to the lowering of the column adaptor. These results show very good agreement with the anticipated state of degradation of the resin and suggest the usefulness of looking beyond the most obvious part of the chromatogram, i.e. the elution phase, that can be affected by column degradation. All three phases studied allow detection of the change in chromatographic performance and may thus be used for monitoring.

3.3. PLS modeling of chromatography data

A PLS regression was used to model the column degradation. Since PLS is a supervised machine learning method and requires a target variable, the cycle number was used as a prediction goal. This is clearly a superfluous task; the cycle number is known and not a subject to statistical uncertainty or noise. The rationale behind this approach is to help the PLS model extract from the chromatograms the signals corresponding to the "regular" aging of the resin (reflected approximately in the cycle number). As an extension of this idea, a large discrepancy between the predicted and the actual cycle number can be indicative of an unexpected or atypical degradation of the column. Thus, a significant error in the predicted cycle number is a strong indicator of changes potentially affecting the column performance and the product quality.

The size and the sequential nature of the dataset constrains the training and validation methods that can be used to fit and benchmark the model. To mimic an approach that can be applied for monitoring actual columns, the first *n* cycles can be used to train the model, which will then be utilized to predict the remaining *30-n* cycles. Note that the underlying assumption is that the initial *n* cycles are normal, i.e., exhibit typical aging. In the subsequent analysis *n=10* was used. Thus, the test set contains both normal and harsh CIP cycles, which can be used to compare the predictions of the model in both cases.

In Figure 10, the absolute values of the prediction errors of the model trained using the initial ten cycles of the *Wash*, *Elution*, and *Strip* phases are shown. The ability of the chromatograms of these three phases to track column aging has been confirmed by analysis of a separate lifetime study dataset and by visual inspection. The column demonstrates consistent aging reflected in a good predictive performance of the PLS model until cycle number 18 (compare with Figures 7-9). Beyond that (for the cycles in which the harsh CIP has been applied several times), significant deviations between the predicted and actual cycle number appear. In fact, the predictions of the model eventually become completely non-sensical, reaching large *negative* values for the cycle number. Similar results are obtained by using chromatograms of individual phases as inputs to the PLS model (data not shown). Thus, the prediction of a supervised model trained on a small number of controlled cycles can be used to detect problems beyond these cycles.

3.4. Similarity Score

The Similarity Score [10] consists of determination of the correlation to a reference chromatogram (cycle 2) and the peak maximum as indicators. Figure 11 shows the correlation of absorbance at 280 nm vs the Peak maximum, indicating that column deterioration caused by the overly harsh CIP can be detected as deviations in the absorbance first by a reduced correlation but over time also as an increase in peak height, consistent with the findings from Principal Component analysis. The *Elution* phase shows the opposite trend compared to the *Wash* and *Strip* phases with respect to peak height because the column deterioration leads to lower peak heights that are detected from around cycle 18. If the ligand degradation leads to weaker binding or overall less binding a lower peak height in elution (and an increased peak height during *Wash* and *Strip* in the case of weaker binding) is expected.

3.5. SOP-MPT

The SOP-MPT used the position of the peak maximum UV absorbance and extracted the corresponding conductivity measurement as the output. The shift in the correlation pattern between the absorbance peak position and the conductivity level starting after the onset of the harsh CIP cycles can again be seen for the *Wash* phase, around cycle 19 as seen in Figure 12. The analysis of the *Elution* and *Strip* phases with the SOP-MPT method did not reveal any patterns related to the column degradation (data not shown).

3.6. Comparison of data analytical tools

The relevant metrics calculated from each data analysis tool were compared with respect to their ability to describe changes in the step yield (see Figure 2). The absolute value of the Pearson correlation between each metric and the step yield across all cycles is calculated and compared to the correlation between cycle number and step yield in Figure 13. A successful metric that extracts predictive information should have a higher correlation than the correlation to the cycle number. There are more successful metrics in the *Wash* and *Strip* phases compared to the *Elution* phase.

4. Conclusions

The potential for using features of the absorbance profiles in selected phases as a real time process analytical tool that indicate deviations in column performance has been investigated. Mathematical tools were able to discern the modification of column performance as early as after a few cycles only of introduction of the harsh CIP exceeding the recommended conditions. Deviations were detected even before the column degradation resulted in reduced step yield and could thus provide an early warning of column degradation.

SOP-MPT was the least capable tool as it was able to detect deviations only in the *Wash*, out of the three phases analyzed. PCA and PLS as well as the Similarity Score were comparable in their predictive capability power: all three phases can be used to extract features that correlate well with a resin being increasingly degraded. Based on the work here described, the user is encouraged to investigate multiple phases in the chromatogram (e.g. *Wash*, *Elution* and *Strip*) for variability and assure that the absorbance signal is within the linear detector range. Afterwards, PCA, PLS and/or Similarity Score can be chosen, as described here, for real time analysis.

The use of multivariate statistics such as PCA or PLS for process monitoring is well known in the literature [1, 22-24]. They are inherently more abstract than for instance a peak height or position, but that is also their merit. They can handle (detect) more complex changes in the process performance and thus potentially become powerful tools. As with any other monitoring approach, outputs should be scrutinized for a consistent trend versus occasional variability of negligible practical impact. As it was observed with PCA data analysis, just lowering the top adaptor of the chromatography column resulted in an “out of trend” signal. Therefore, analyzing a larger dataset can help distinguish between insignificant run-to-run variabilities and trends that have practical impact. This could inform action thresholds for “alert” and “alarm” limits. Conveniently, 3-15 runs, as used on this study, proved to be a good enough set to establish a reference “normal” baseline, using conditions demonstrated during lifetime studies to allow column reuse. PCA and PLS allows you to compare the scores to the Hoteling T2 ellipsis (where the significance level can be adjusted) for deviations in the principal component hyperplane and the DModX critical level for deviations orthogonal to the hyperplane. The settings for such levels should be determined based on the normal operation of the process and cannot be deducted from vendor data. The same approach could be applied to the other methods investigated here but that has been beyond the scope of this proof-of-concept paper.

The benefit of the proposed approach from an implementation perspective is that it doesn't require real-time calculations as all calculations are performed between chromatographic cycles. It also utilizes only existing data, i.e. signals that are already recorded in the historian such as UV, conductivity, pH and pressure. The data processing steps required depends on the data collection format, if the different signals don't have synchronized time steps across cycles one needs to align the data points to a common set of data intervals by linear interpolation between the existing data points. Then the starting point in time for the chromatographic phases of interest should be retrieved and used to align the chromatogram to the phase start. All the proposed methods are then just a simple weighted sum calculation (PCA and PLS), a data look up operation (SOP-MPT) or calculation of the correlation to a reference chromatogram (Similarity Score).

While potential to use these methods to anticipate column health was demonstrated, the approach still requires further work before it can be established as a PAT. Additional points of interest include the investigation of different scales and molecules. Follow-up work will include the analysis of column lifetime studies using "normal" chromatography methods to determine limits of performance in routine commercial batch production. The findings described here will be contrasted with the more subtle transitions of these datasets but still expected to correlate with the health or cycle number of the column bed. Furthermore, it will be useful to understand whether the degradation mode of the resin is similar in a gentler chromatography method. Finally, it would be very useful to find out commonalities in chromatography-based processes, for example, by having defined phases to analyze regardless of the feed or molecule. The application of the concept presented here could be of particular interest for the monitoring of continuous chromatography processes. In this case, if chromatography profiles are to transition atypically, product streams can be diverted before other performance (or product quality) issues onset.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Fig 1

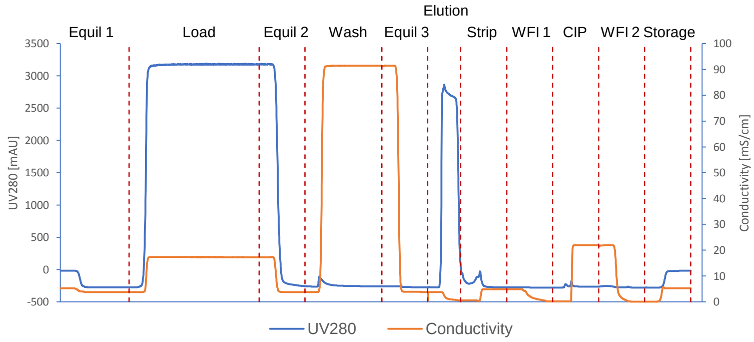


Figure 1.Typical protein A capture chromatogram with the phase changes indicated by red dashed lines. The traceline for absorbance at 280nm reaches saturation level in the Load phase and in the Elution. The buffer used in Equil 1, Equil 2 and Equil 3 was Tris-based and had neutral pH. The Wash used a high NaCl content buffer also at neutral conditions. Elution used an acetate-based buffer at low pH. The resin was regenerated with a combination of acidic buffer (Strip) and NaOH solution (CIP).

Fig 2

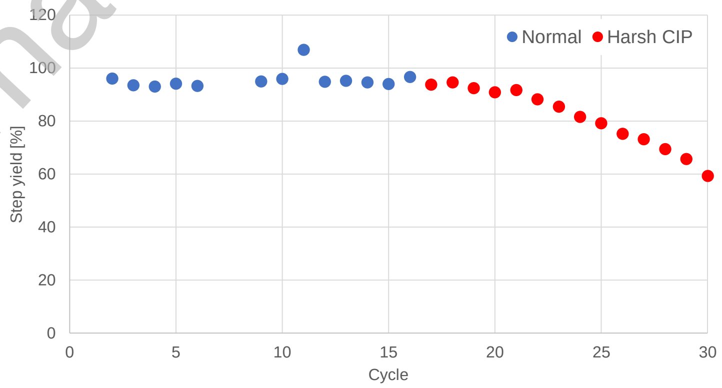


Figure 2. Step yield measurements over the first 15 chromatography runs using NaOH exposure as recommended by the vendor followed by 15 additional runs which used an harsh CIP.

Fig 3

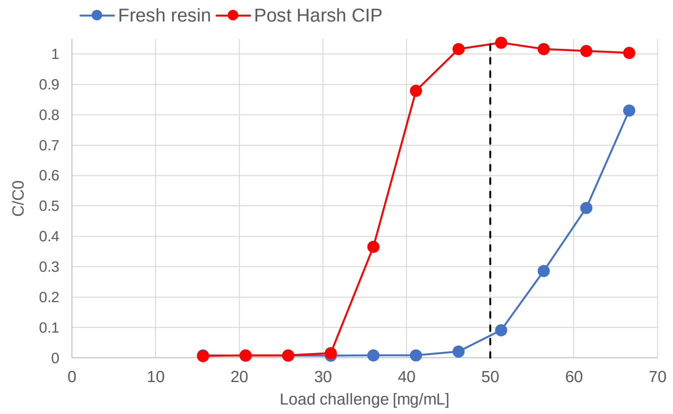


Figure 3. Dynamic binding capacity for fresh resin (pre-run) and after the harsh CIP (post-run). The black dashed line indicates the load challenge during the study (50 mg/mL).

Fig 4

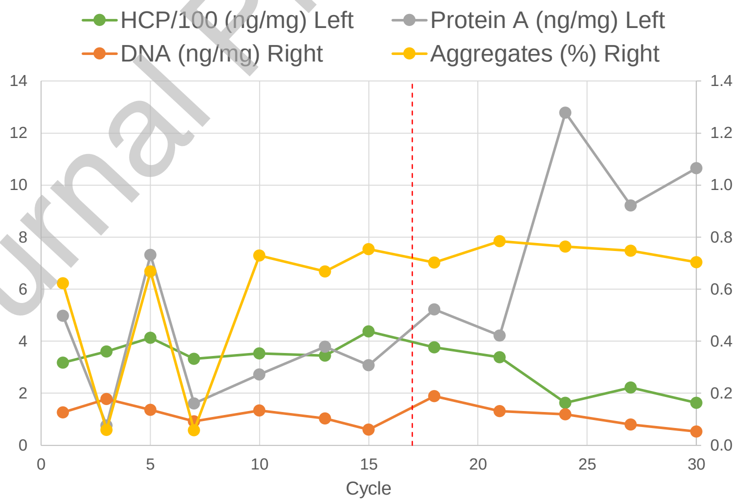


Figure 4. Product quality for selected chromatography cycles during the normal cycles (1-16) and the harsh CIP cycles (17-30). The red dashed line indicates the change from normal to harsh CIP.

Fig 5

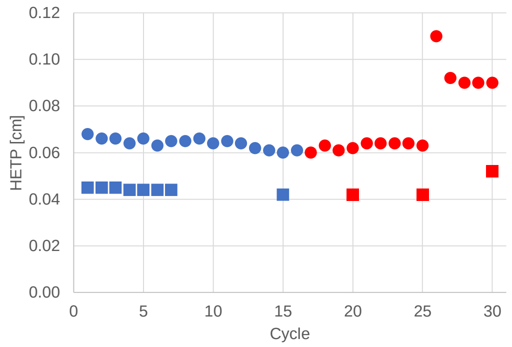


Figure 5. Comparison of HETP from pulse test (squares) and transition analysis (circles). The normal CIP cycles are indicated by blue symbols and the harsh CIP by red symbols.

Fig 6

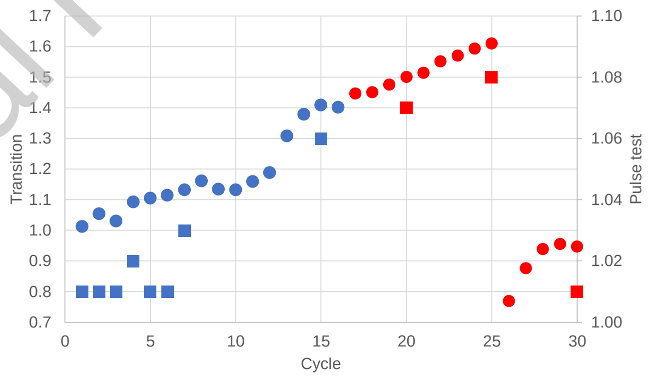


Figure 6. Comparison of asymmetry data from pulse test (squares) and transition analysis (circles). The normal CIP cycles are indicated by blue symbols and the harsh CIP by red symbols.

Fig 7a

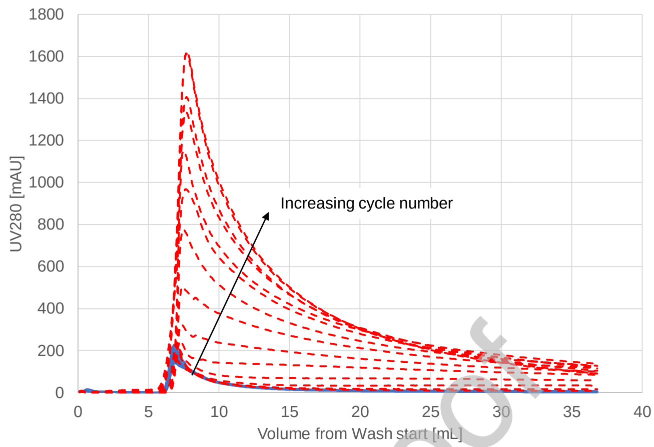


Fig 7b1

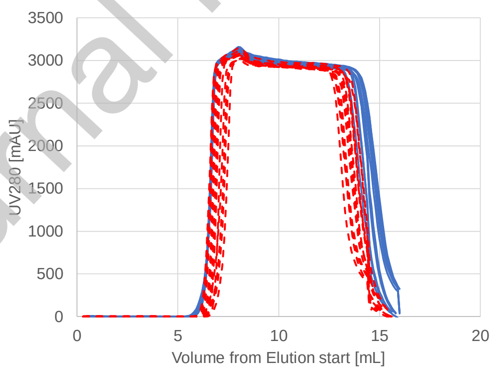


Fig 7b2

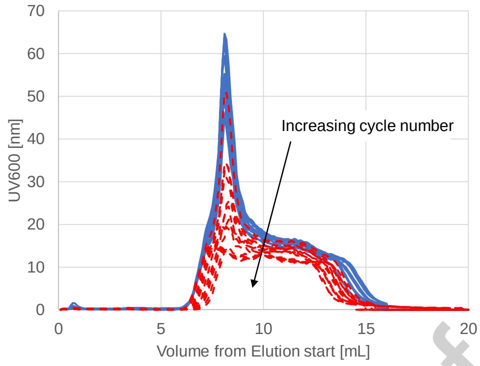


Fig 7c

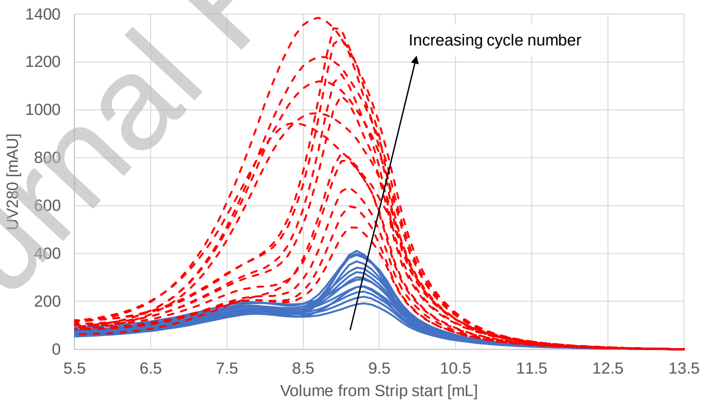
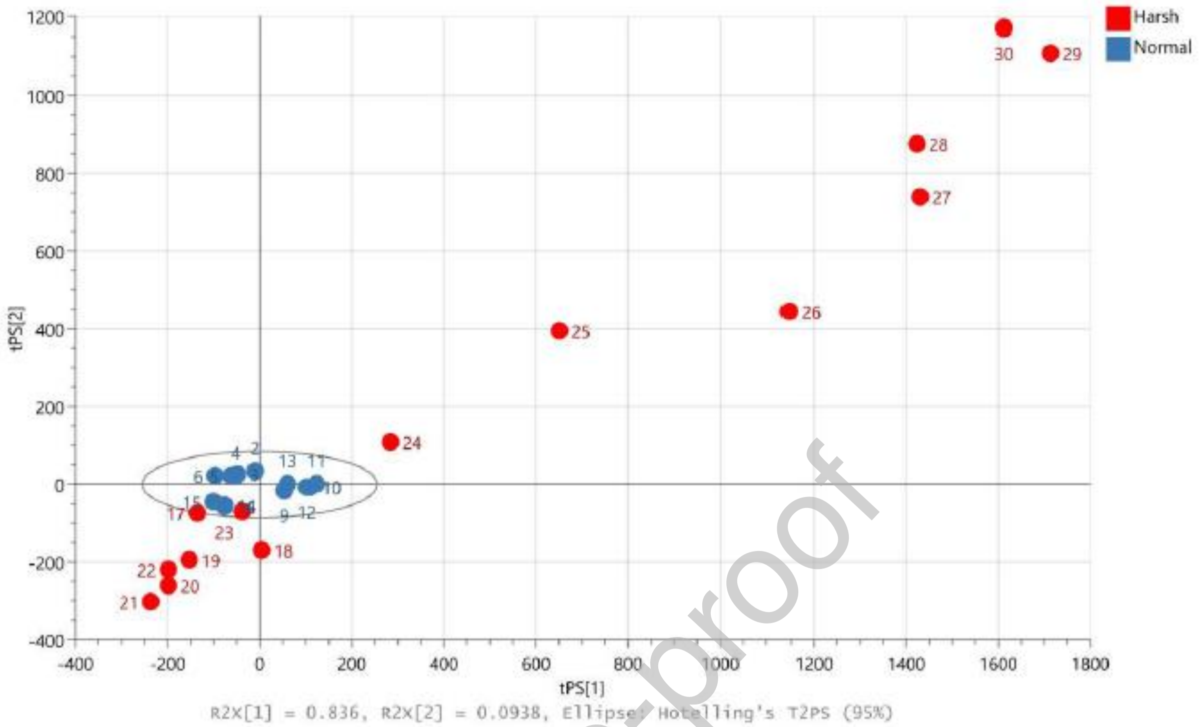
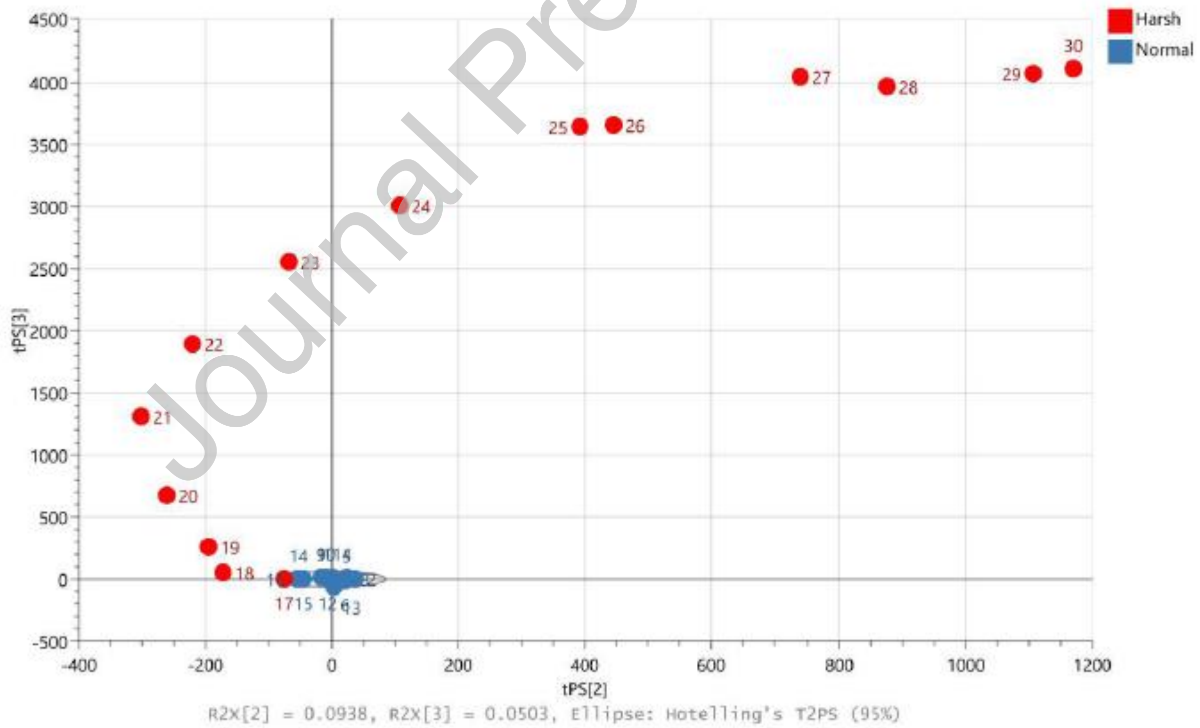


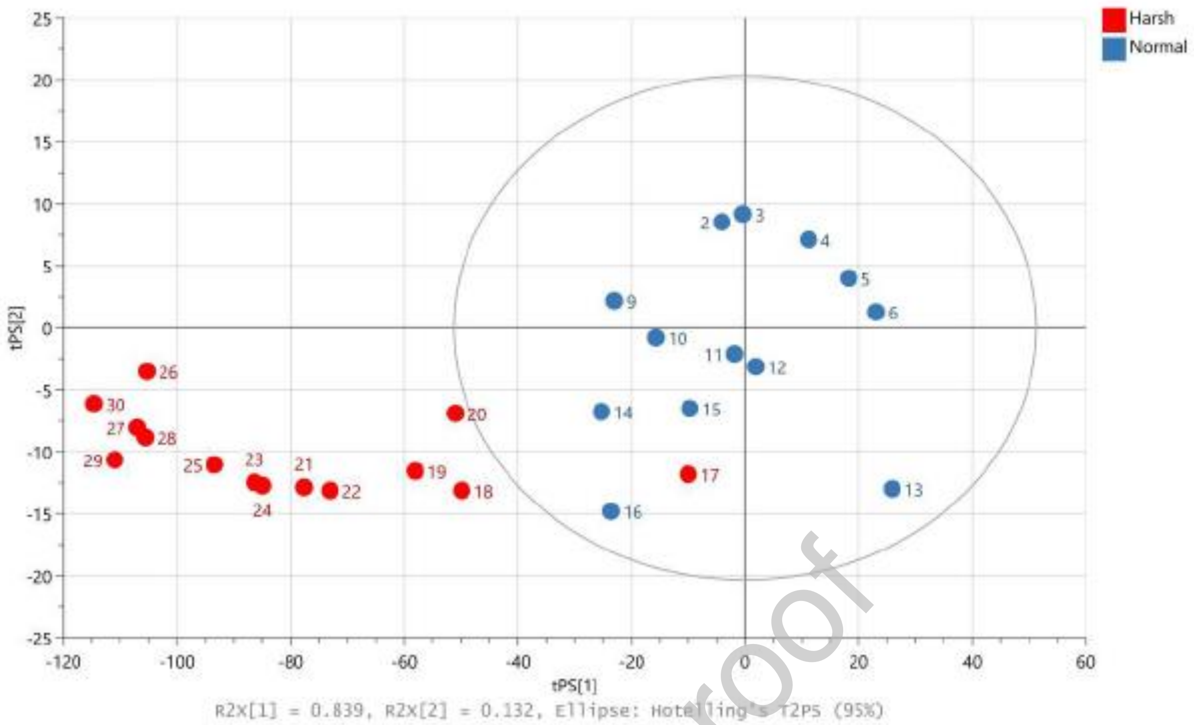
Figure 7. Absorbance chromatograms of the Wash phase at 280 nm (a), Elution phase measured at 280 nm (b1) and 600 nm (b2), and Strip phase at 280 nm (c). Cycles post harsh CIP exposure (17-30) are shown in red dashed lines.



Harsh
Normal
tPS[2]
tPS[1]
R2X[1] = 0.836, R2X[2] = 0.0938, Ellipse: Hotelling's T2PS (95%)



Harsh
Normal
tPS[3]
tPS[2]
R2X[2] = 0.0938, R2X[3] = 0.0503, Ellipse: Hotelling's T2PS (95%)



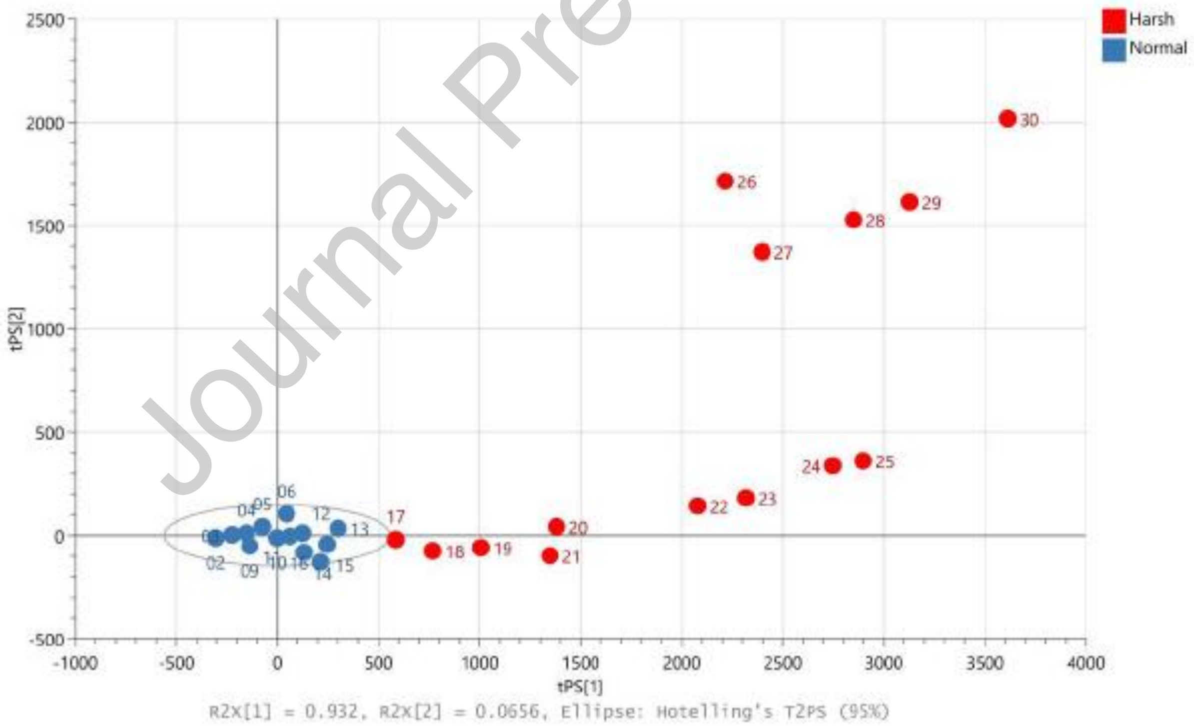


Figure 8. PCA score plot based on model training using the first 15 cycles with the remaining cycles projected. Wash (a1 shows PC1 vs PC2, a2 shows PC2 vs PC3), Elution at 600 nm (b) and Strip (c). Run 16 is included with the “normal CIP” runs since the harsh CIP only takes place at the end of the chromatography method. tPS designates the predicted scores for each principal component and R2X the proportion of the total variance explained per component.

Fig 9a

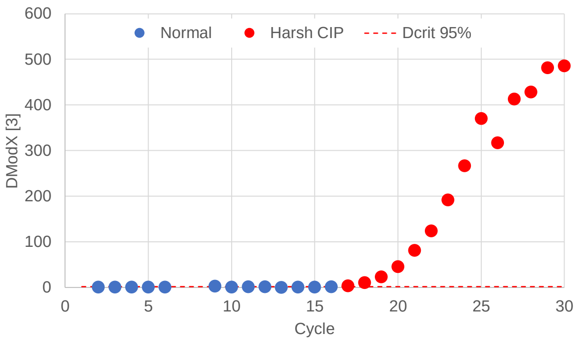

Normal
Harsh CIP
Dcrit 95%
DModX [3]
600
500
400
300
200
100
0
0
5
10
15
20
25
30
Cycle

Fig 9b

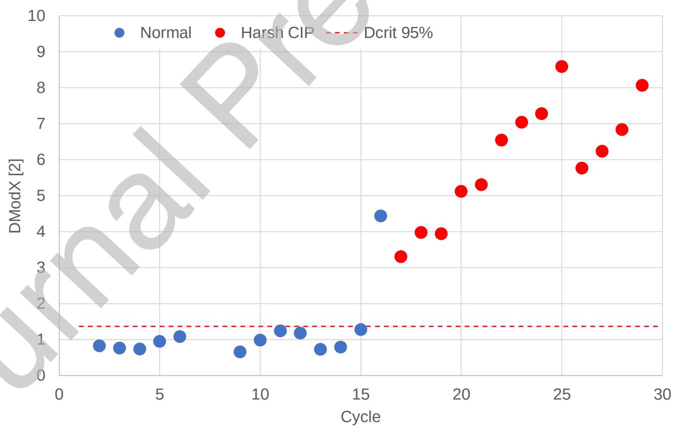

Normal
Harsh CIP
Dcrit 95%
DModX [2]
10
9
8
7
6
5
4
3
2
1
0
0
5
10
15
20
25
30
Cycle

Fig 9c

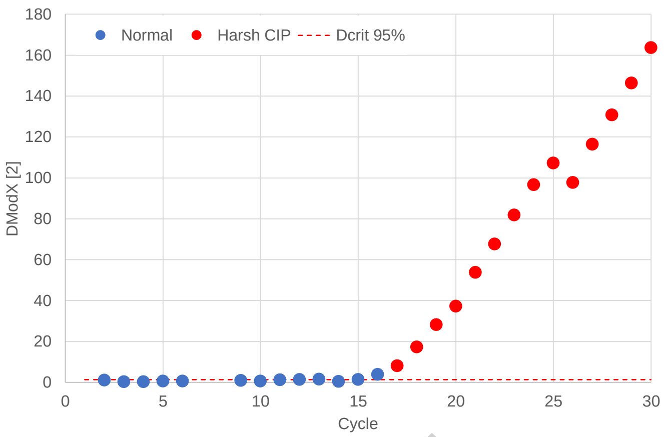


Figure 9. Distance to model plot for the a) Wash, b) Elution at 600 nm and c) Strip. Cycle 16 is included with the "normal CIP" cycles since the harsh CIP takes place at the end of the chromatography method. Dcrit is 1.371.

Fig 10

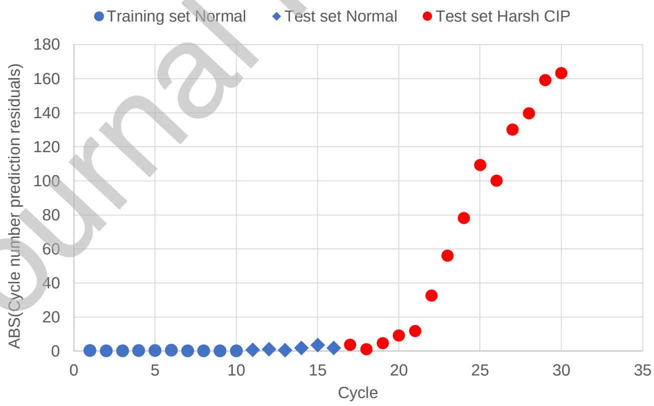


Figure 10. The results of supervised modeling: a PLS model was trained to predict the cycle number, with the initial 10 cycles used as a training set, and the remaining 20 cycles as a test set. The predictions are consistent with the actual cycle number until cycle 18, after which they become progressively worse as shown by the absolute values of the prediction error.

Fig 11a

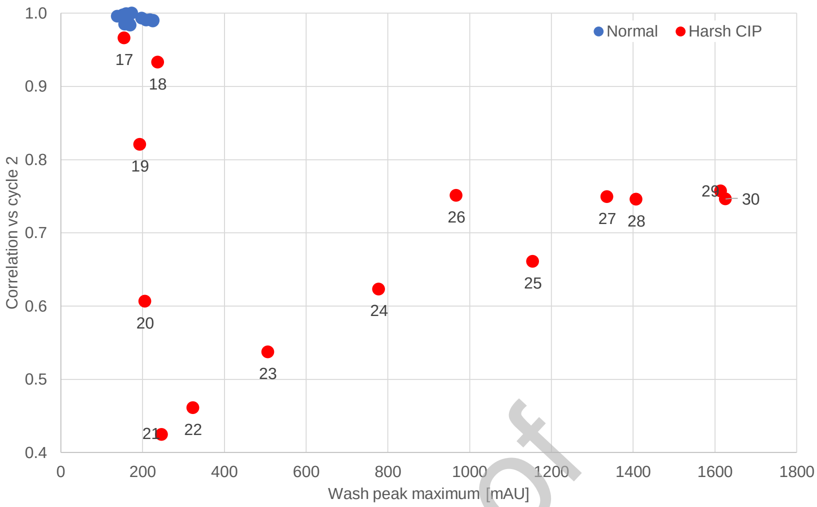


Fig 11b

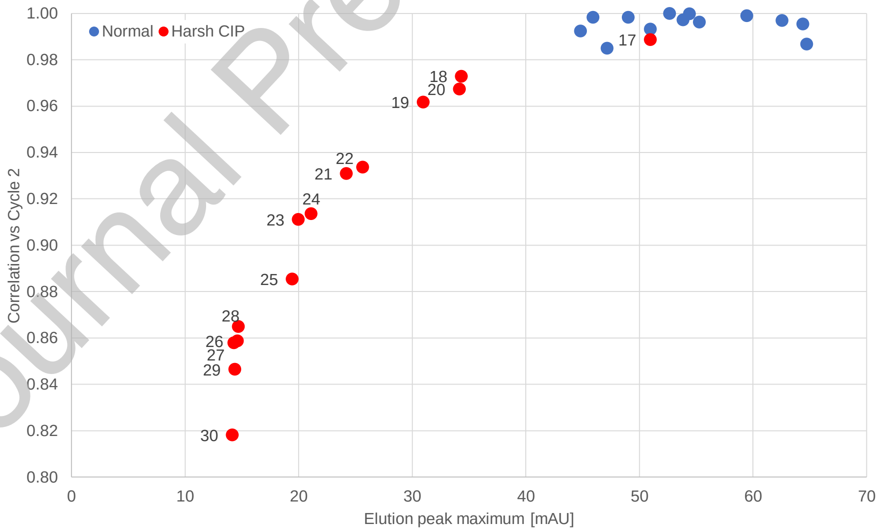


Fig 11c

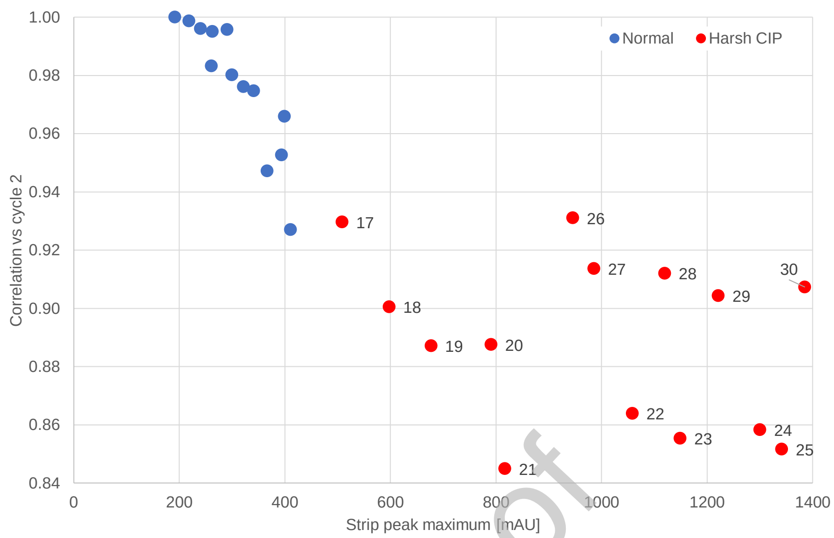


Figure 11. Similarity Score plot with cycle 2 as reference, for a) Wash, b) Elution at 600 nm and c) Strip. Cycle 16 is included with the "normal CIP" cycles since the harsh CIP takes place at the end of the chromatography method.

Fig 12

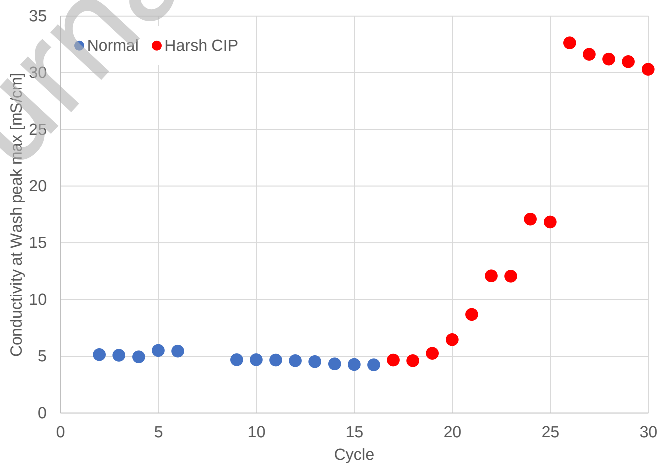


Figure 12. SOP-MPT analysis of the Wash phase. Outlier cycles 1,7 and 8 excluded.

Fig 13

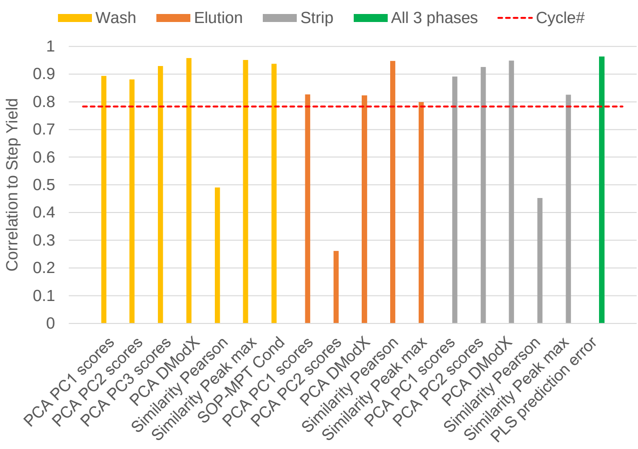


Figure 13. Comparison of different metrics with respect to the absolute value of the correlation to step yield over the cycles. Only the relevant metrics for each phase are shown. The correlation to Cycle number is used as a cutoff criteria to indicate an improved predictive ability. The metrics are the calculated PCA scores for PC1, PC2 (and PC3 in Wash), PCA distance to model (DModX), similarity score Pearson correlation and peak max, SOP-MPT for conductivity and absolute value of the PLS cycle number prediction error.